

EE236 LAB-2

16 August 2024

1 Aim of the experiment

- To find forward voltage, reverse saturation current and ideality factor of the given PIN diode (Infineon BAR 15-1).
- To find reverse recovery time of the given PIN diode at various frequencies and compare with the that of normal PN junction Diode.
- To observe the working of PIN diode as an RF switch at different DC bias voltages.

2 Design

The circuit consists of a constant voltage source V_{dc} of 5V, and a $1k\Omega$ potentiometer to vary voltage across the diode, and a 100Ω resistor.

For part-2, a square pulse voltage of $2 V_{pp}$ is given. The circuit is simple, with a diode and a $1k\Omega$ resistor connected in series with the voltage source. The reverse recovery time is measured at the voltage frequencies 10 KHz, 100 kHz, 1 MHz and 3 MHz

We made the circuit as given in the problem statement, consisting of a diode, two $100nF$ capacitors, a $6 V_{ptop}$ sine wave of 10MHz as voltage source, two 500Ω and one 50Ω resistor and a V_{bias} Voltage. The V_{out} and I_{diode} for both the PIN diode and PN Junction Diode is measured at different V_{bias} voltages: 5 V, 0 V, 1 V, 3 V and 5 V.

3 Experimental Results

3.1 I-V characteristics of PIN Junction

We had to measure V_d and I_d for PIN diode. Below are the observations and graphs of I_d vs V_d and $\ln(I_d)$ vs V_d :

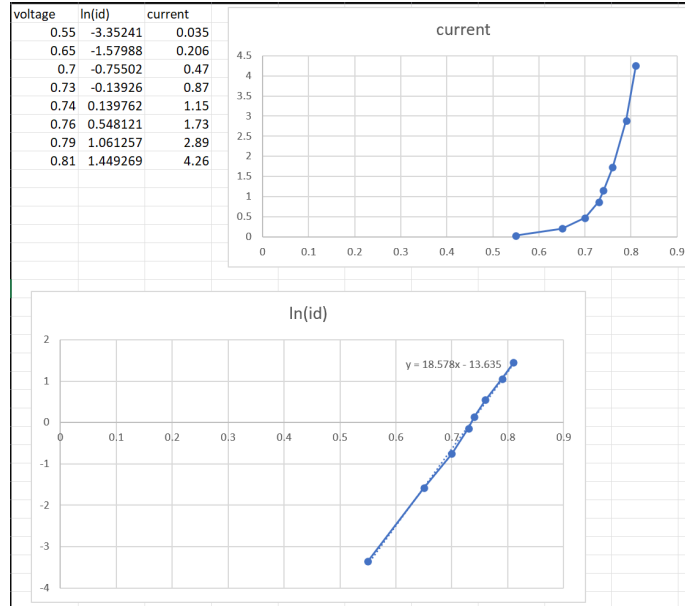


Figure 1: I-V characteristics

From the above observations, we note that the slope and intercept of $\ln(I_d)$ vs V_d graph is 18.578 and -13.635 respectively. Using the equations from the previous lab, we get reverse saturation current and ideality factor as:

$$I_s = e^{y-intercept} = 1.19 \times 10^{-3} mA \quad (1)$$

and

$$\eta = \frac{1}{slope \times 0.026} = 2.07 \quad (2)$$

respectively. For cut-in Voltage, we consider the Voltage at which current is 1mA. So the cut-in voltage is 0.73V approximately.

3.2 Reverse Recovery Time of PN junction and PIN Junction

Frequency	RRT of PIN	RRT of PN
10 kHz	1.25 μs	1.76 μs
100 kHz	1.04 μs	1.06 μs
1 MHz	485 ns	337ns
3 MHz	154 ns	159ns

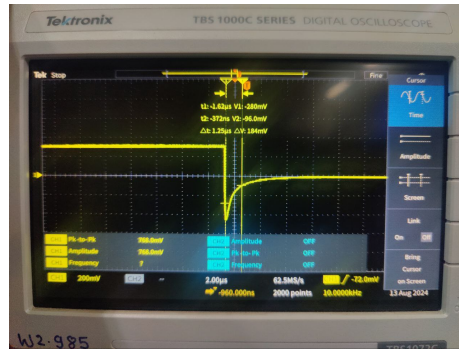


Figure 2: PIN 10kHz



Figure 3: PIN 100kHz



Figure 4: PIN 1MHz

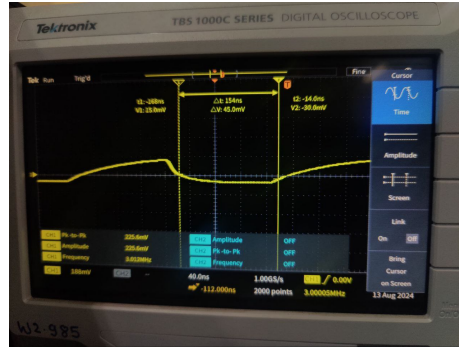


Figure 5: PIN 3MHz

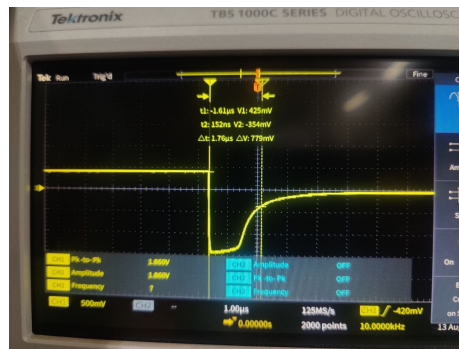


Figure 6: PN junc 10kHz



Figure 7: PN junc 100kHz

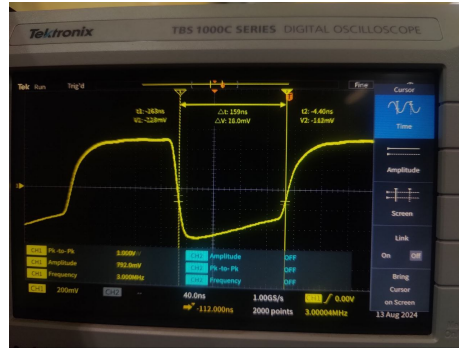


Figure 8: PN junc 3MHz

3.3 RF Switch Circuit of PN junction and PIN Junction

V_{bias}	$I_d(PIN)$	$V_{pp}(PIN)$	$I_d(PN)$	$V_{pp}(PN)$
-5V	0.001 mA	34 mV	0.001 mA	88 mV
0V	0.207 mA	234mV	1.34 mA	648mV
1V	0.726 mA	472mV	0.62 mA	556mV
3V	2.24 mA	540mV	1.64 mA	564mV
5V	4.21 mA	548mV	3.40 mA	572mV

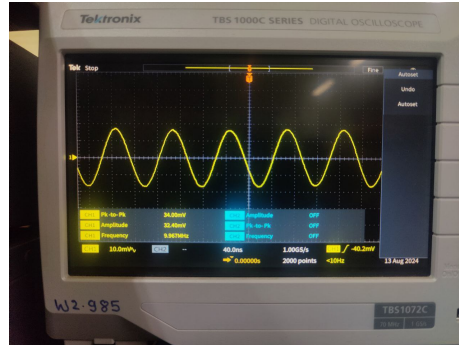


Figure 9: PIN -5V

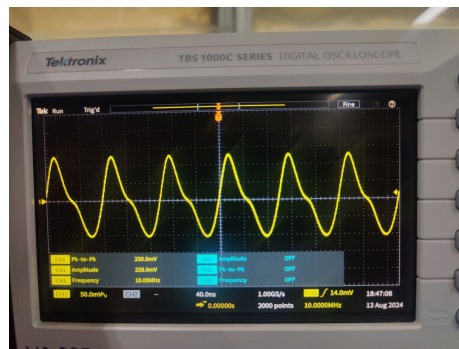


Figure 10: PIN 0V

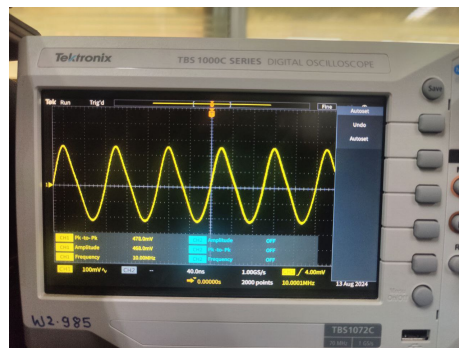


Figure 11: PIN 1V

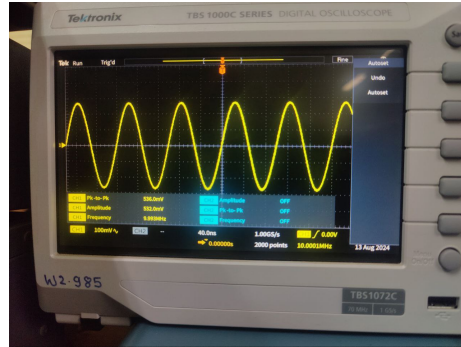


Figure 12: PIN 3V



Figure 13: PIN 5V

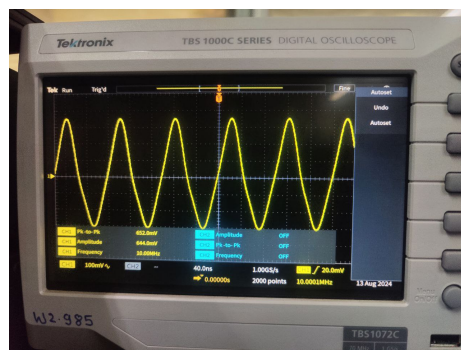


Figure 14: PN 0V

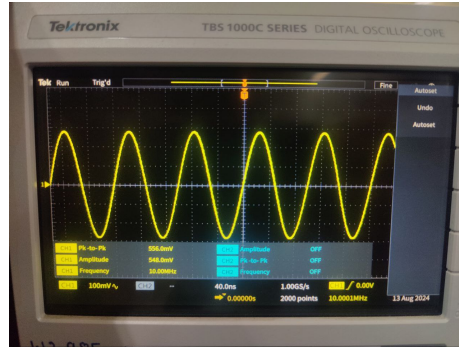


Figure 15: PN 1V

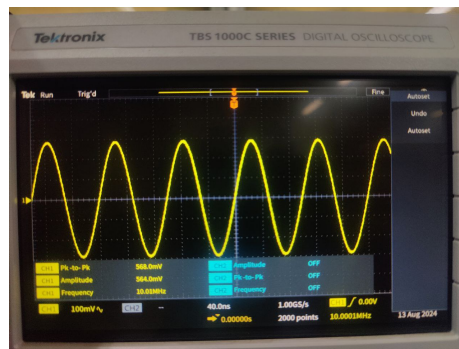


Figure 16: PN 3V

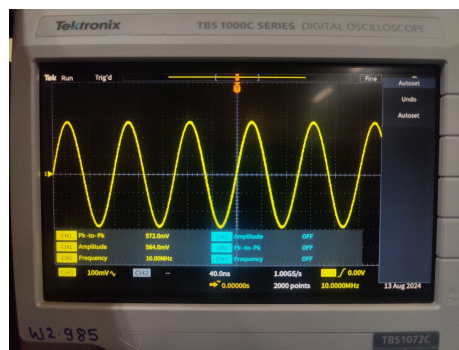


Figure 17: PN 5V